

EEE4861 Assignment

Design and Investigation of an Ultra-Low-Power CMOS Food-Freshness Decision Circuit for Biodegradable Smart Packaging

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Motivation

Food spoilage during storage and transportation remains a major challenge, particularly in developing countries where continious cold-chain monitoring is often unavailable. Smart food packaging can improve food safety by monitoring environmental conditions and providing timely warnings before food becomes unsafe for consumption. However many existing smart packaging systems depend on microcontrollers or wireless communication modules, increasing power consumption, manufacturing cost; and electronic waste. A simple low-power CMOS hardware decision circuit integrated with biodegradable packaging offers a practical and sustainable alternative for disposable food-packaging applications.

Objective and CEP statement

This assignment aims to design and investigate a **low-power CMOS hardware decision circuit** intended for smart food packaging using biodegradable packaging materials.

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✓ The proposed circuit is designed using E and Electronic Engineering principles and analyzed using OrCAD Capture / Pspice.

✓ A comprehensive literature review is conduc on low-power AI hardware circuits and er -mentally sustainable smart packaging tech

✓ suitable biodegradable packaging materials evaluated, and the most appropriate ma is selected based on technical and er -mental considerations.

✓ A practical method is proposed for integ the CMOS hardward decision circuit wit biodegradable food packaging to monitor fo freshness

✓ The proposed design considers food safety environmental sustainability, and the obje of Bangladesh. Environmental Conservation ECP 2023.

Critical challenges

Designing a smart food-packaging system involves several engineering challenges. The circuit must consume extremely low power while maintaining reliable operation using a minimal number of transistors. At the same time, it should be suitable for integration with biodegradable packaging materials without significantly increasing manufacturing cost. The design must also support practical food-quality monitoring while minimizing environmental impact.

Conflicting Requirements

- Achieving ultra-low power consumption while maintaining reliable logic operation
- Integrating electronic circuitry with biodegradable packaging materials without affecting food safety
- Maintaining low manufacturing cost while improving packaging intelligence
- Designing a simple hardware decision circuit that performs reliable food freshness monitoring without complex processing hardware.

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1. Course Outcomes (COs) : EEE 4861 (VLSI Desig

CO4 : Engage in independent research on emer
topics in VLSI and low-power hardware de

CO5 : Consider food safety, environmental sust
-ty and biodegradable packaging in engineer
design.

2. Program Outcomes (POs) : EEE 4861 (VLSI Desig

PO3 (Modern Tool Usage) : Create, select and
apply appropriate engineering techniques,
simulation tools and modern design metho
to solve complex engineering problems

PO7 (Environment and Sustainability) : Underst
and evaluate the environmental and socie
impact of engineering solutions while pro
sustainable development.

3. Complex Engineering Problems (Ps)

P1 : The project requires knowledge of CMOS d
circuit design, low-power VLSI techniques,
smart food-packaging technologies

P2 : Conflicting design requirements exist bet
low power consumption, low manufacturing
reliable operation, and environmental sust

P3: There is no single predefined hardward architecture for Intelligent food-freshness monitoring The designer must justify an appropriate low-power hardware decision circuit based on engineering analysis.

P4: The design requires consideration of food safety, biodegradable materials, sustainable packaging, and environmental regulations beyond convention -al CMOS circuit design

4. Knowledge Profiles (Ks)

✓ The proposed work demonstrates engineering knowledge corresponding to K3-K8 including CMOS circuit design, simulation, literature review, sustainability assessment, and engineering decision-making.

CO-PO-K-P-Mapping

COs	POs	knowledge Profile(s)	CEP	CEA
CO4	PO5	K6	P1, P2, P3	—
CO5	PO7	K7	P2, P4	—

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Abstract

Smart food packaging has emerged as an e approach for improving food safety by continuo monitoring storage conditions and providing ea warning of spoilage. However, many existing smart packaging systems rely on microcon wireless communication modules, or complex artificial intelligence (AI) processors, which i power consumption, manufacturing cost and electronic waste. This work proposes the and electronic waste. This work proposes design and simulation of a compact low-p CMOS hardware decision circuit for smart foo packaging using biodegradable materials. Th proposed circuit employs a two-input CM NAND gate to process binary warning signa generated by threshold-based gas and tem sensors, producing a simple edge decision re food freshness

The circuit was designed and simulated or Capture/Pspice using a 3.3V supply. Simu results verified correct NAND logic operat where the output remained HIGH for input combinations 00 01 and 10 and changed when both sensor warnings were activel representing a critical food spoilage cond Current and transient analysis showed negli

static current, an average power consumption of approximately 10- 15nW, and a propagation delay of about 100ps, demonstrating the suitability of the design for ultra-low-power applications. To improve environmental sustainability, several biodegradable packaging materials were evaluated using a weighted decision matrix. Polylactic acid (PLA) was selected as the most suitable substrate because of its good mechanical strength, food-contact safety, printability, and compatibility with electronic integration. The proposed design combines low-power CMOS hardware with biodegradable packaging to reduce energy consumption, minimize electronic waste, and support environmentally sustainable smart food-packaging applications.

To improve environmental sustainability, several biodegradable packaging materials were evaluated using a weighted decision matrix. Polylactic Acid (PLA) was selected as the most suitable substrate because of its good mechanical strength, food-contact safety, printabity, and compatibility with electronic integration. The proposed design combines low-power CMOS hardware with biodegradable packaging to reduce energy consumption, minimize electronic waste, and support environmentally sustainable smart food-packaging applications.

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C1] Introduction

1.1) Low-power AI hardware is increasingly used i devices where sensing and decision making be performed with limited energy and a area. For dispasable smart packaging, complex processors and continious wireless commu are often unnecessary. A compact CMOS circuit can instead process thresholded sen signals locally with low power and minimo hardware.

This project investigates a two-input CMO circuit for food-freshness monitoring. Gas temperature warning signals are used as inputs and a critical condition is identifi when both warnings are active. Alongside t circuit design, biodegradable packaging are reviewed to identify a suitable sub for environmentally safe and sustainable implementation.

1.2) Literature Reveiw 1

Paper Title: Hardware Implementation of Deep Net Accelerators Towards Healthcare and Biomedica Applications

Authors: M.R Azghadi et al

Key findings

- Edge AI accelerators can process data locall without continiously transferring it to cloud
- CMOS, FPGA and emerging-memory platforms

reduce latency and communication energy.

- Hardware acceleration is useful for low-power sensing and classification applications.
- Circuit architecture must be selected according to power, accuracy and computational requirements

Limitations

- Neural-network accelerators generally require more hardware than simple rule-based classifiers
- Accuracy, power consumption and silicon area involve design trade-offs.
- Complex architectures may be unsuitable for low cost disposable packaging.

Literature Review 2

Paper Title: Recent Advance of Intelligent Packaging Aided by Artificial Intelligence

Authors: X. Li et al

Year: 2023

Key findings

- Intelligent packaging can combine sensors, indicators and AI-based analysis for food-freshness detection.
- Temperature, gas concentration and chemical changes are useful indicators of food condition.
- Localized data processing can improve real-time monitoring and reduce dependence on manual inspection.

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- AI-assisted packaging may reduce food waste improve supply-chain frequency.

Limitations

- Many systems require complex sensing, data processing or communication hardware
- Sensor calibration and environmental variatio affect prediction reliability.
- Cost and large-scale manufacturing remain important barriers.

Literature Review 3

Paper Title: A Review on Gas Indicators and Senso Smart food Packaging.

Authors: W. Heo et al.

Year : 2024

Key Findings:

- Oxygen, carbon dioxide and ammonia indic can provide information about food condit
- Gas sensors can detect changes associa package leakage, microbial activity and sp
- Sensor selection depends on sensitivity, d range, response method and the type of packaged food
- Gas sensing is suitable for developing fres warning systems in intelligent packaging

Limitations

- Different food products release different gases, so a single sensor is not suitable for every application.
- Humidity and temperature may influence sensor response
- Practical implementation requires calibration and protection from direct food contact.

Literature Review 4

Paper Title: A Physically Transient Form of Silicon Electronics

Authors: S W Hwang et al.

Year : 2012

Key findings:

- Functional electronic components can be fabricated using materials designed to dissolve or disappear after their operating life
- Transient electronics can reduce persistent electronic waste in temporary and disposable applications.
- The work established an important foundation for environmentally degradable electronic systems

Limitations :

- Fabrication methods are more complex than conventional CMOS manufacturing.
- Controlled degradation depends strongly on the selected materials and environmental conditions.

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- A conventional silicon die attached to biodegra packaging cannot automatically be considered biodegradable.

Literature Review 5

Paper Title: A Review on Bio-Based Polymer Polyla Acid Potential on Food Packaging.

Authors: Ds Rajendran et al.

Year : 2024

Key Findings:

- PLA is renewable, biodegradable and suitable fo several food-packaging applications.
- It offers good transparency, processability c mechanical strength.
- PLA can be manufactured as films, container composite packaging structures.
- Blending and reinforcement can improve its and mechanical performance.

Limitations

- Neat PLA can be brittle.
- Its heat resistance and moisture/gas barrie properties are lower than those of some petroleum-based plastics.
- Effective degradation normally requires cont composting conditions.
- Modifications may increase production cost

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Literature Review Summary

The reviewed studies show that low-power hardware can perform localized sensing and decision-making with less communication and processing overhead. Smart food-packaging research supports the use of gas and temperature information for freshness assessment, while transient electronics studies demonstrate the environmental value of reducing persistent electronic waste. PLA is identified as a promising biodegradable packaging material, although moisture resistance, brittleness and composting requirements must be considered.

Based on these findings, this project uses a simple CMOS NAND circuit rather than a complex neural-network accelerator. The circuit performs a fixed two condition hardware decision with only four transistors, making it more suitable for low cost disposable packaging.

1.3 Selection of Biodegradable Packaging Material

Proposed Material: PLA based biodegradable packaging film.

The packaging material must provide sufficient mechanical strength, moisture resistance, food-packaging suitability and compatibility with electronic integration. Four candidate materials were considered: PLA film, coated paper, thermoplastic starch and

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polyhydroxyalkanoate (PHA).

PLA offers a balanced combination of surface q mechanical stability, processability and biodegradabi Paper is inexpensive and easily available but req additional moisture-barrier coatings. Thermoplastic is highly biodegradable but absorbs moisture PHA provides excellent biodegradability and bar performance but remains relatively expensive less widely available. General reviews of biode food-packaging materials identify these same tr offs among barrier performance, strength, h resistance and cost

Comparison of Candidate Biodegradable Packagi Materials

Property	PLA	PHA	Coated Paper	Ther star
Mechanical strength	Good rigidity and dimensional stability	Good flexibility and toughness	Moderate; weak when exposed to moisture	Low mech stre
Moisture Resistance	Good	Very good	Poor without protective coating	Poor to wate abso
Food-contact Suitability	Widely used in food packaging	Suitable for food packaging	Suitable after coating	Suita for food appli

Property	PLA	PHA	Coated Paper	Thermoplastic starch (TPS)
compatibility with Electronics	Smooth surface suitable for sensor and CMOS integration	Suitable but requires more complex processing	Printable, but moisture affects reliability	less suitable because of poor dimensional stability
Commercial Availability	Widely available	Limited and relatively expensive	Widely available	Moderate availability
Environmental Sustainability	Renewable and biodegradable	Biodegradable and bio-based	Recyclable and biodegradable	Biodegradable but less durable.

Justification of Material Selection

PLA is selected as the packaging material because it provides the best overall balance of **mechanical strength, moisture resistance, food-contact safety, and compatibility with electronic integration.** Its smooth and stable surface is well suited for mounting the proposed CMOS hardware decision circuit and associated sensors, while its widespread commercial availability makes it practical for large scale smart packaging application.

Although PHA offers comparable biodegradability and better moisture resistance, its higher cost and limited availability reduce its practicality. Coated paper and thermoplastic starch are environmentally

friendly alternatives, but their lower moisture resist and structural stability make them less suitable for reliable electronic integration. Therefore, PLA is considered the most appropriate biodegradable packaging material for the proposed low-power CMOS hardware decision circuit.

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Chapter 2: LOW POWER CIRCUIT DESIGN AND TOOLS USAGE.

2.1 Application of EEE knowledge in Circuit Diagram.

The work applies fundamental Electrical and Electronic Engineering concept to design a low-power CMOS decision circuit for smart food packaging. The circuit is implemented using CMOS logic, where two PMOS transistors from pull-up network and two NMOS transistors from pull-down network, PSpice simulation is used to verify the logic operations, where Microwind is used to represent the physical implementation through stick and layout designs.

2.2 Proposed CMOS Circuit Design

The proposed hardware consist of a two input CMOS NAND gate operating from a 5V supply.

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The gate receives two digital inputs representing gas and temperature sensor conditions and produces a food freshness decision. The circuit contains four mosfets, two NMOS and two PM

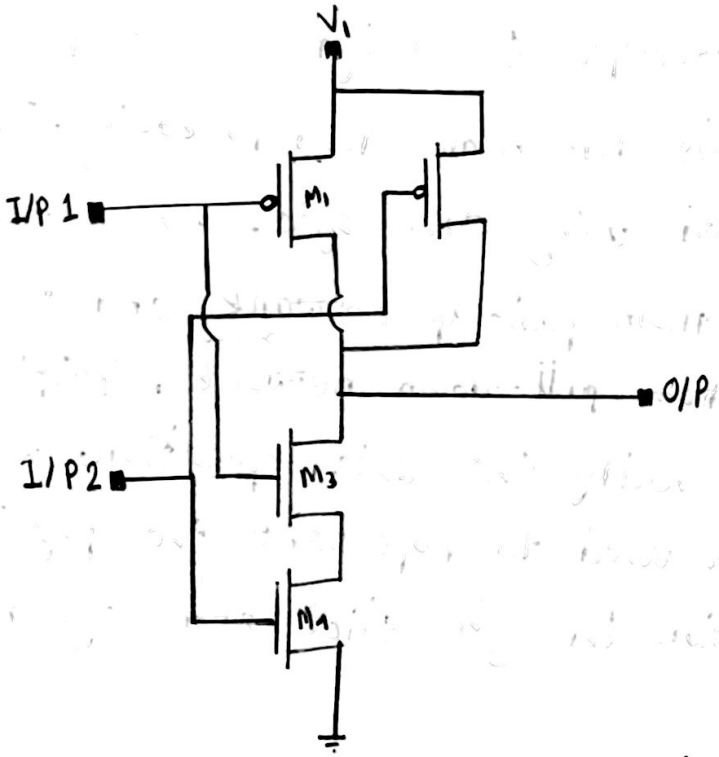


Fig. Proposed low power CMOS logic circuit.

Here, I/P 1 : warning from a gas or spoilage related sensor.

I/P 2 : Warning for temperature or environment sensor.

logic low represent → normal sensor condition

logic high → corresponding threshold has been exceed

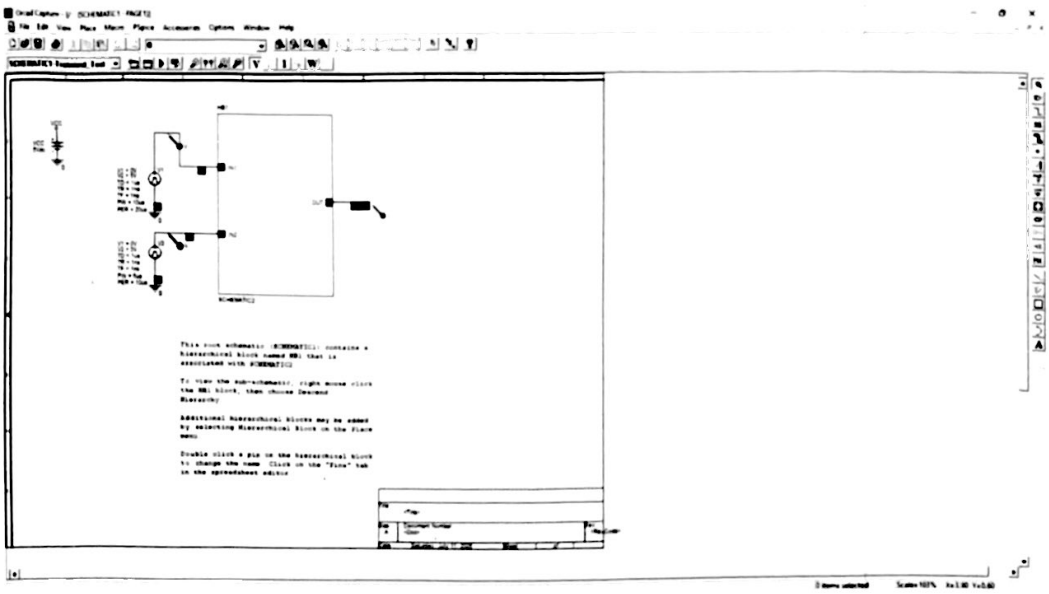


Fig. : Top-level Orcad capture schematic showing two i/p sources, hierarchical CMOS NAND block and O/P node.

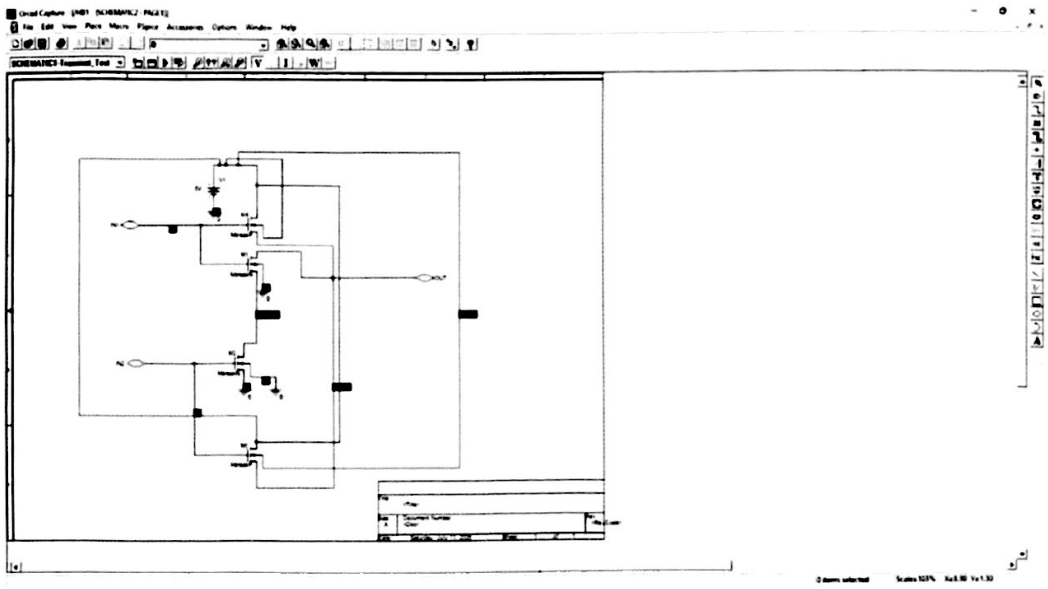


Fig. : Transistor level implementation of two i/p CMOS NAND decision circuit.

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2.3] Pspice Simulation

The designed circuit was verified using transient analysis in Pspice. Two VPULSE source was applied to generate all possible output conditions (00, 01, 10, 11). The simulation confirmed that output remained high except when both inputs were High, which matches the NAND gate.

I/P 1	I/P 2	OUT	Food monitoring interpretation
0	0	1	normal
0	1	1	temperature - caution
1	0	1	gas - caution
1	1	0	critical condition

For simulation,

Logic function : 2 I/P CMOS NAND

No. of transistor : 4

PMOS : two in parallel

NMOS : two in series

supply voltage : 5V

i/p low : 0V

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i/p high : 5 V

o/p low : approx 5 V

o/p high : approx 0V

simulation duration : 40μs

i/p rise time : 1ns

i/p fall time : 1ns

max simulation step : 10ns

target power range : nW – μW

Measured avg. power : ~(10-15) nW

Mesaured propagation delay: ~ 0.1 ns

The design is quite simple because food quality condition usually changes slowly. Therefore, a comp processor or high freq digital accelerator was not required for initial decision function.

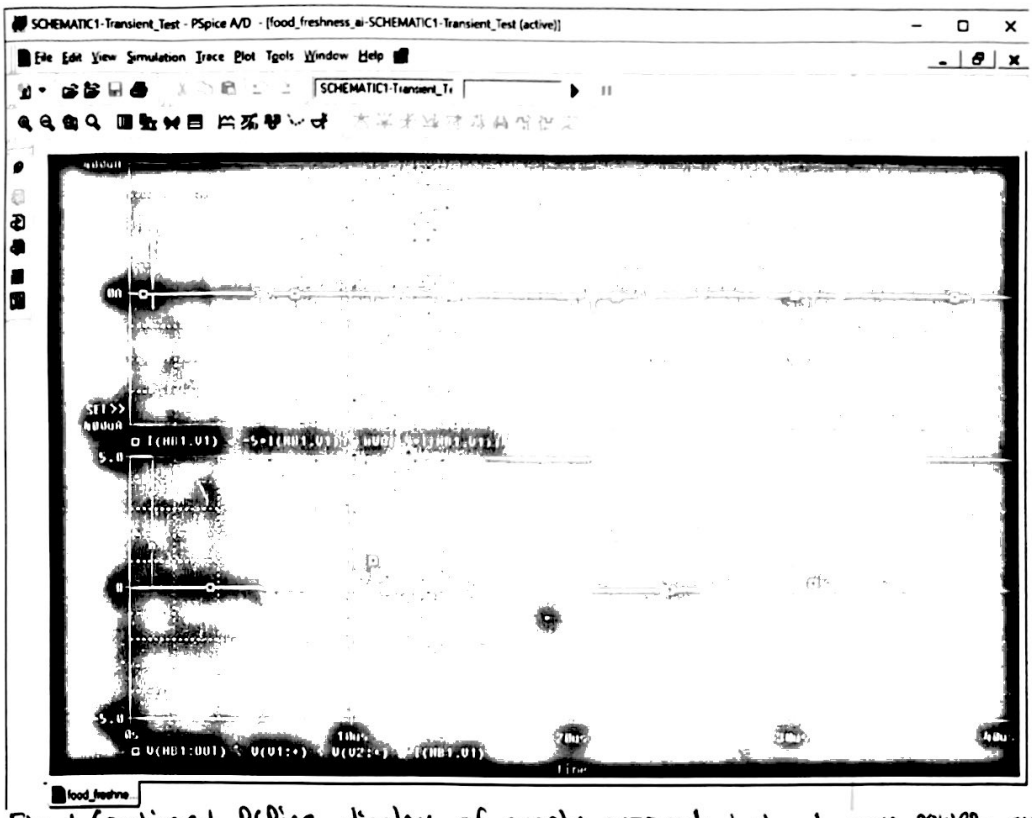


Fig.: Combined PSpice display of supply current, instantaneous power, avg power.

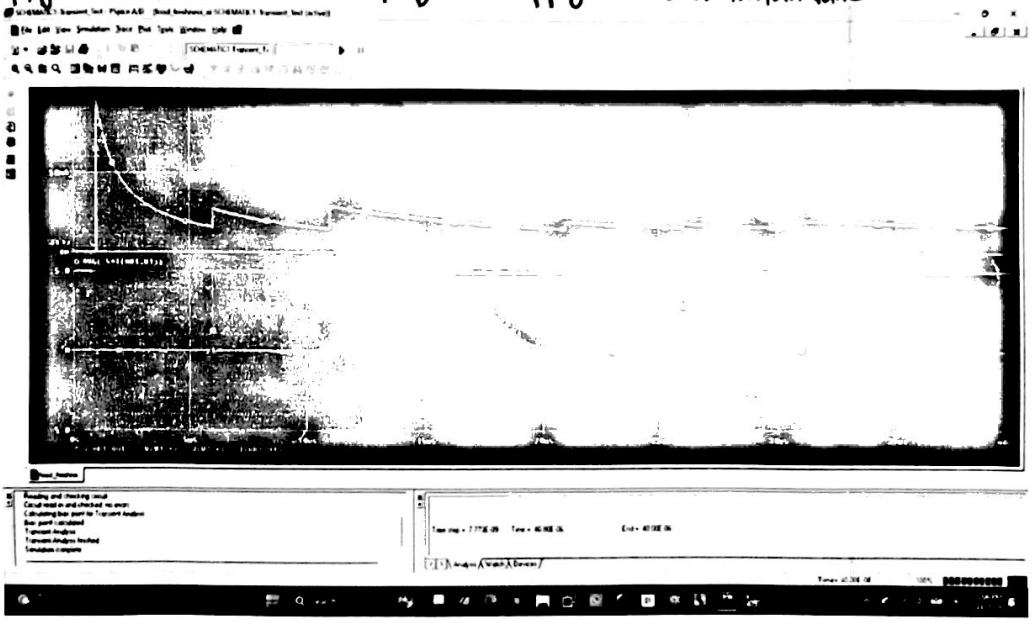


Fig.: Avg. power consumption of CMOS NAND circuit over the transient simulation

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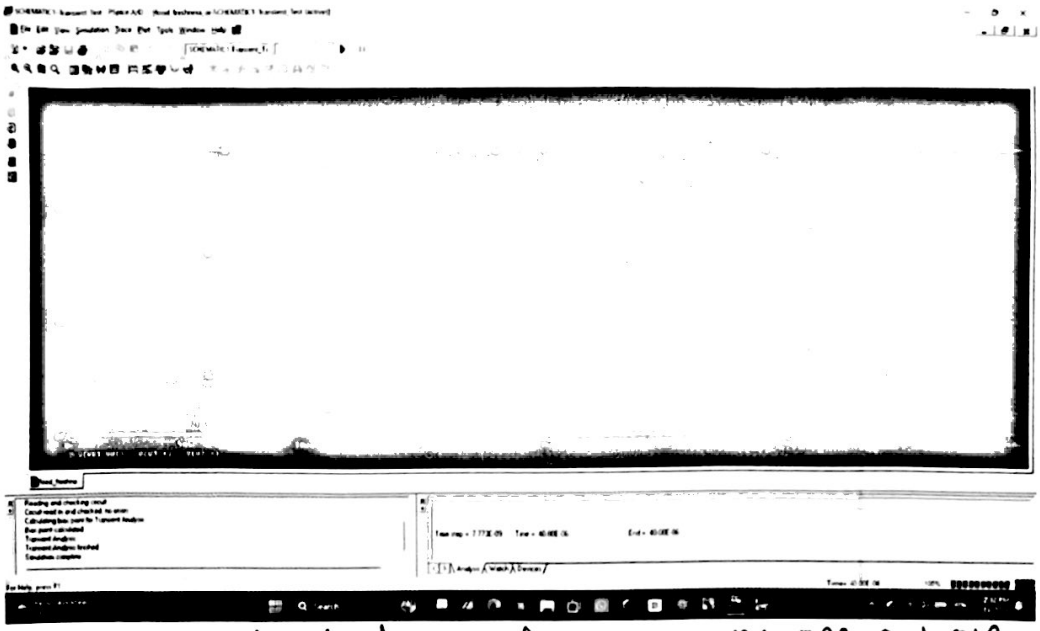


Fig.: Pspice transient waveforms of I/P1, I/P2 and O/P

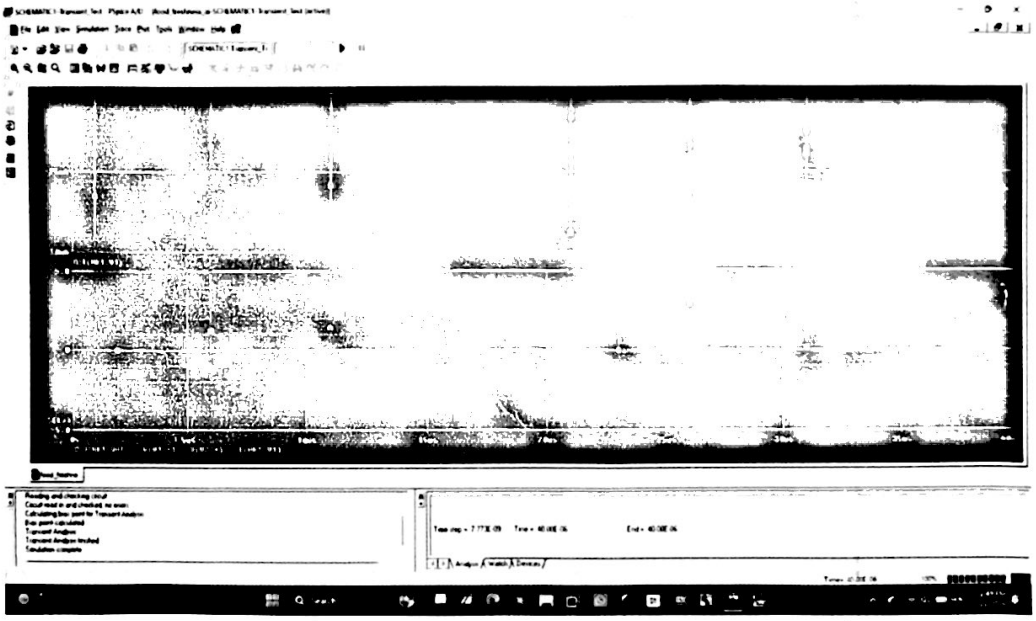


Fig.: Supply current waveform of CMOS NAND circuit during transient operation.

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2.4] Stick diagram and Layout design.

The stick diagram provides a simplified representation of the physical CMOS implementation. It illustrates the arrangement of polysilicon, diffusion region, metal interconnections, VDD and GND while maintaining the same connectivity as the transistor level circuit. The representation helps reduce layout complexity before physical implementation.

The layout design of CMOS NAND gate was generated using Microwind. The layout follows standard CMOS design rule with PMOS transistor placed inside the n-well, NMOS transistor in the p-substrate, Metal-1 routing for power and signal connections and polysilicon gate forming the transistor channel. The layout preserve the same electrical connectivity as the schematic and stick diagram.

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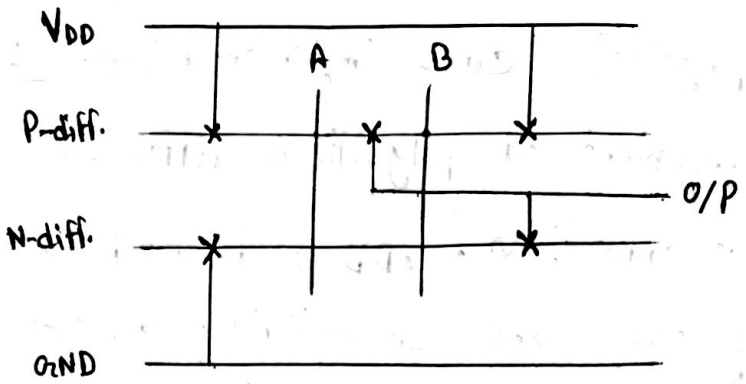


Fig. : Stick diagram

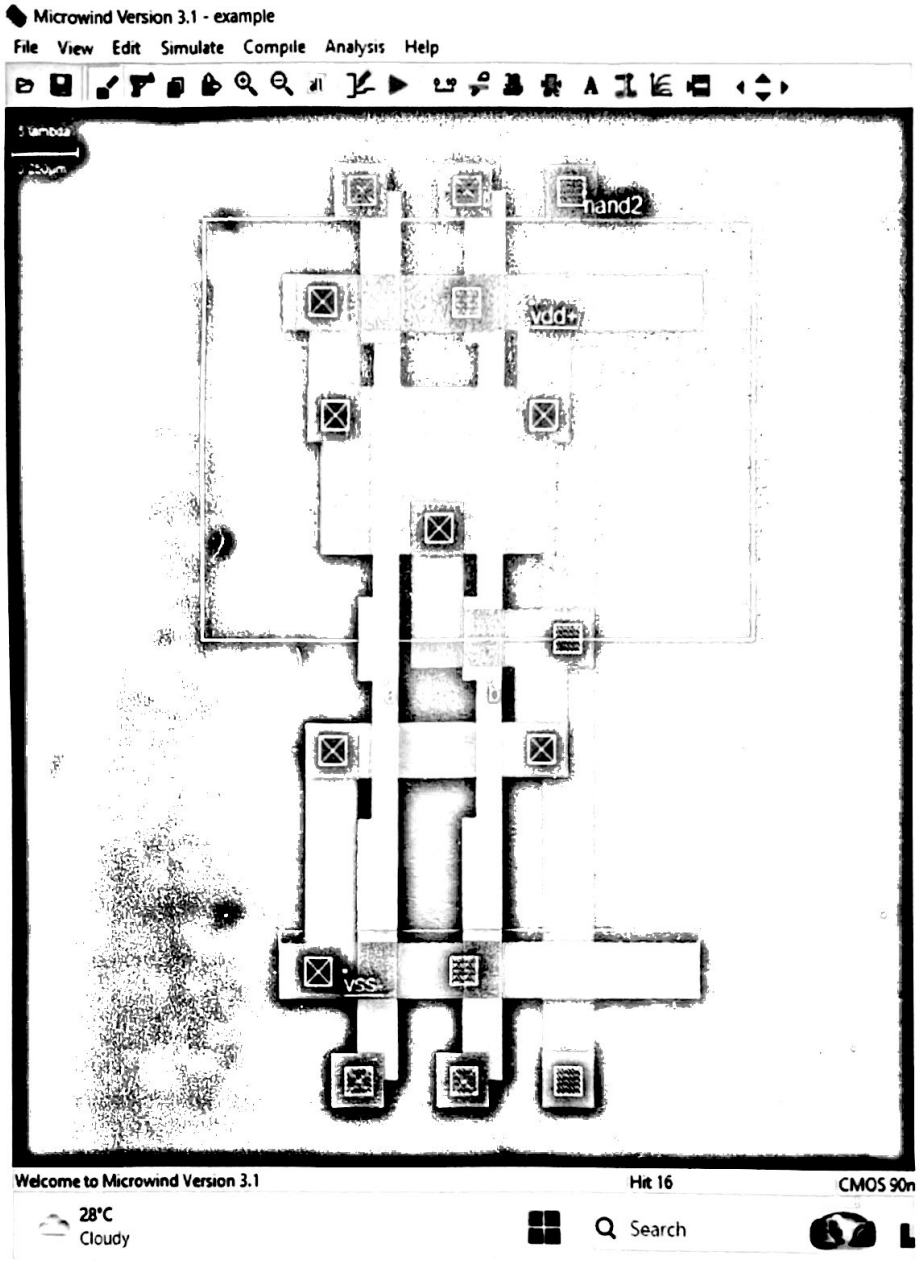


Fig.: Layout design of proposed CMOS

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2.5 Performance Analysis Using SPICE

Transient simulation verified the correct logic operation of the proposed CMOS NAND gate. The circuit exhibit negligible static current, switching current only during logic transitions, an average power consumption of approximately 10-15 nW and a propagation delay of approx. 0.1 ns (100 ps). This result demonstrate that the proposed circuit is suitable for ultra-low -power smart packaging applications.

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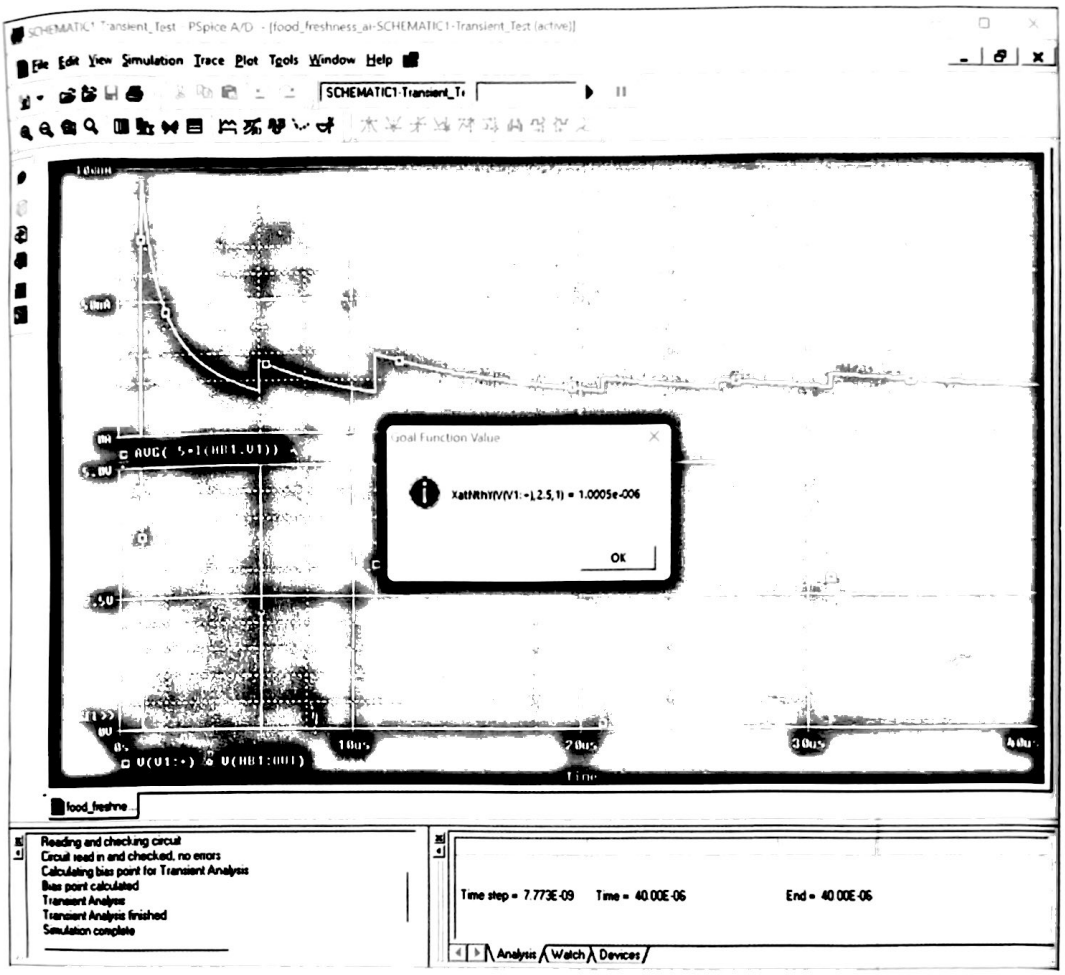


Fig.: PSpice goal-function, evaluation of first 2.5V crossing in i/p1

The average power,

$$P_{avg} = AVG\,[-5I\,(HB1.V1)]$$

at the end of 40 µs simulation,

average power was appox, $P_{avg} \approx 10\text{-}15\ nW$

So, this result places the simulated circuit in nanowatt power range.

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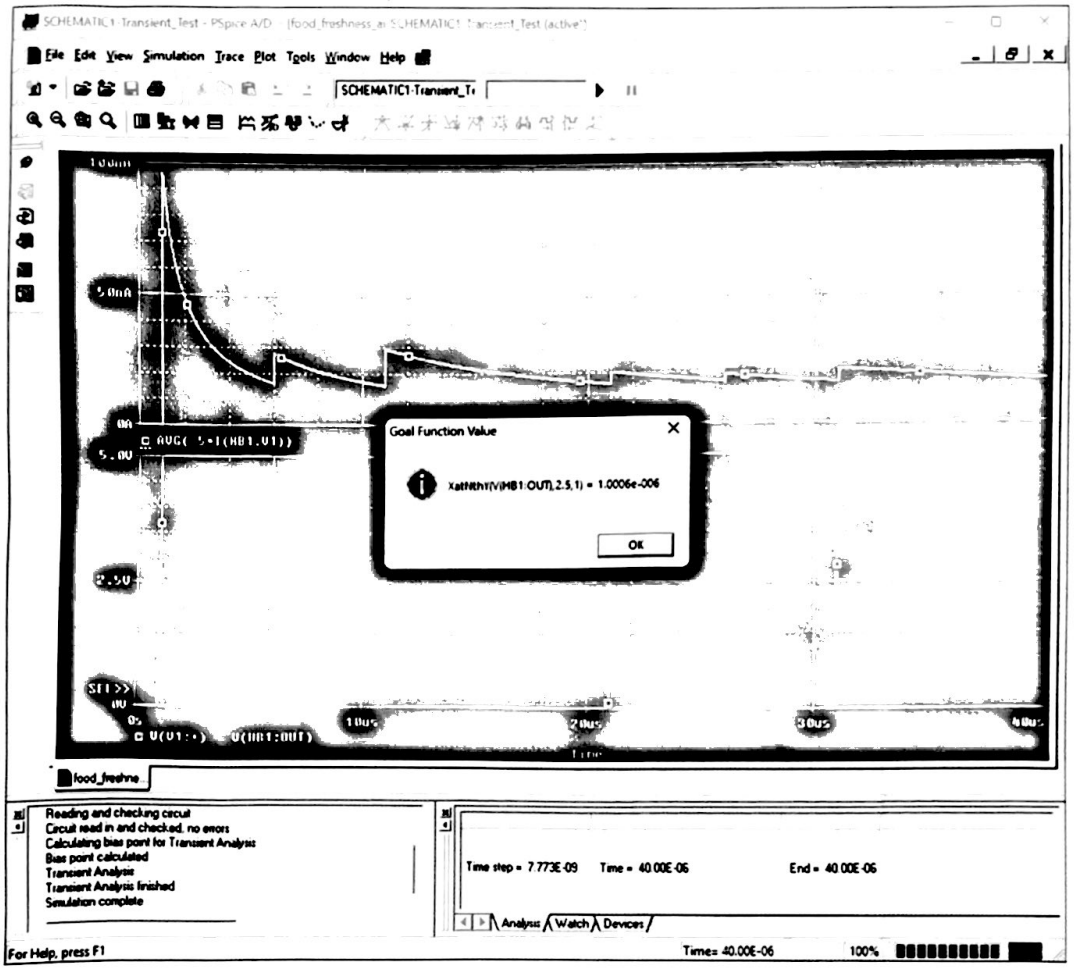


Fig.: PSpice goal function, evaluation of first 2.5V crossing of O/P

Here,

The measured i/p crossing time $t_{IN} = 1.0005 \times 10^{-6}$ s

$\therefore t_{IN} = 1.0005\ \mu s$

The measured o/p crossing time $t_{OUT} = 1.0006 \times 10^{-6}$ s

$\therefore t_{OUT} = 1.0006\ \mu s.$

So,

Propagation delay $t_p = t_{out} - t_{in}$

$= (1.0006 - 1.0005) \times 10^{-6}$ s

$= 1 \times 10^{-10}$ s

$\therefore t_p = 0.1$ ns

$= 100$ ps.

The measured delay is very small because the simulation uses idealized MOSFET and does not include large o/p load or layout parasitics.

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